

Question: What is the gender of your condor chick?

A photograph of a gel electrophoresis result. The gel is oriented vertically with a scale on the left side marked from 0 to 3 cm. On the right side, there is a label 'sinemilione' with an arrow pointing to the right. A single, prominent band is visible in the second lane from the left, and this band is circled with a thick black line. Other lanes show faint bands or no bands at all.

Gel electrophoresis is a method for sorting DNA fragments fragments by size and charge
condor # 32
Well # 4 10 M
Result: male
(male ♂ or female ♀)

Electrophoresis

electrolytic

DNA → change wells

- end negative
Gel
- shorter + end

phoresis - combination of transport (one organism gets a ride then) & travel (separate)

(comes off)

Shorter (faster) the DNA the faster it travels

- female = 2 bands
- male = 1 band

Human Y-chrom Length of piece

	Human	Y-chrom	Length of piece
female ♀	X X	Z W	263,829,0 bp
male ♂	X Y	Z Z	263 bp

yy determines if male

male = ZZ = 263 bp
female = ZW = 263,829,0 bp

Analysis:

1. What are the chromosomes for a female condor?

The chromosomes for a female condor are ZW.

2. For a male condor?

The chromosomes for a male condor are ZZ.

3. Can you tell the gender of condors apart by looking at them?

No you can't tell the gender of them by looking at them because they're monomorphic.

4. About how many bases long is the gene we are looking at on the z chromosome?

We are looking at about 263 bases long on the z chromosome.

5. About how many bases long is the gene we are looking at on the w chromosome?

We are looking at about 290 bases long on the w chromosome.

6. Which chromosome band should travel further down the gel? Why?

The Z chromosome should travel further than the W because Z is shorter (263 bp) than W (290 bp) and shorter DNA fragments travel faster and further.